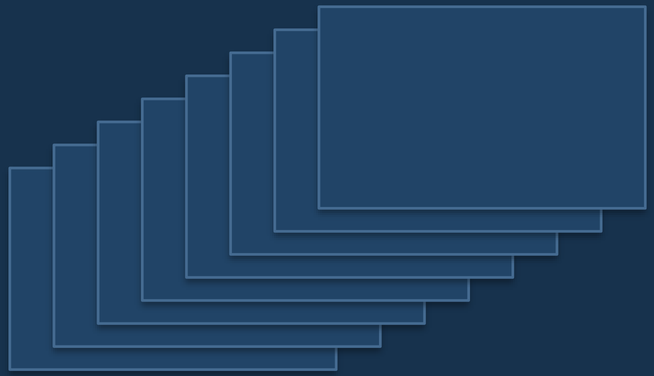
CMP4221 · Multimedia Systems



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Multi-Image Encryption with DWT + Chaotic Map

Compression + encryption in one pipeline

Xu, Gao, Cao, Mou · ACM TOMM Vol. 22 Issue 3 · March 2026
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Thesis

DWT shrinks each color image, the LL sub-bands are stacked into a cube, and chaos scrambles the cube into a secure cipher object.

Sending color images is slow AND unsafe

The paper is solving two costs at once: transmission size and confidentiality.

Bandwidth

A 512×512 RGB image has 786,432 channel values. A batch of images gets heavy fast.

Confidentiality

If images are intercepted, the visible content must be destroyed before transmission.

Multi-image scale

One-image-at-a-time encryption is awkward for batches like medical scans or surveillance frames.

Goal: compress AND encrypt multiple images in one pass.

Ref: Section 1, Introduction

Two ingredients: DWT and chaos

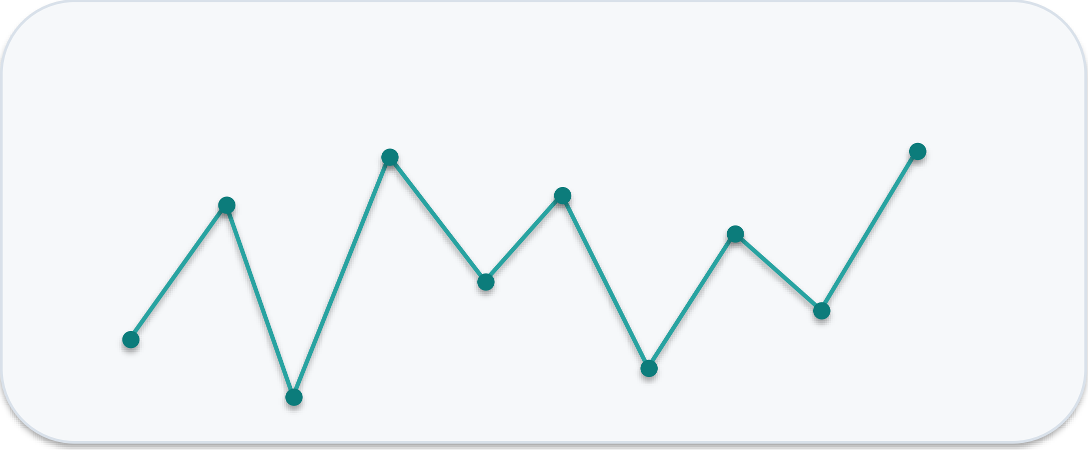
Abdul · 1:30 · Background

DWT compression

LL approximation keep this	**LH** horizontal edges
HL vertical edges	**HH** diagonal edges

Keep LL only → half width × half height = 1/4 size

3D chaotic pseudo-random signal



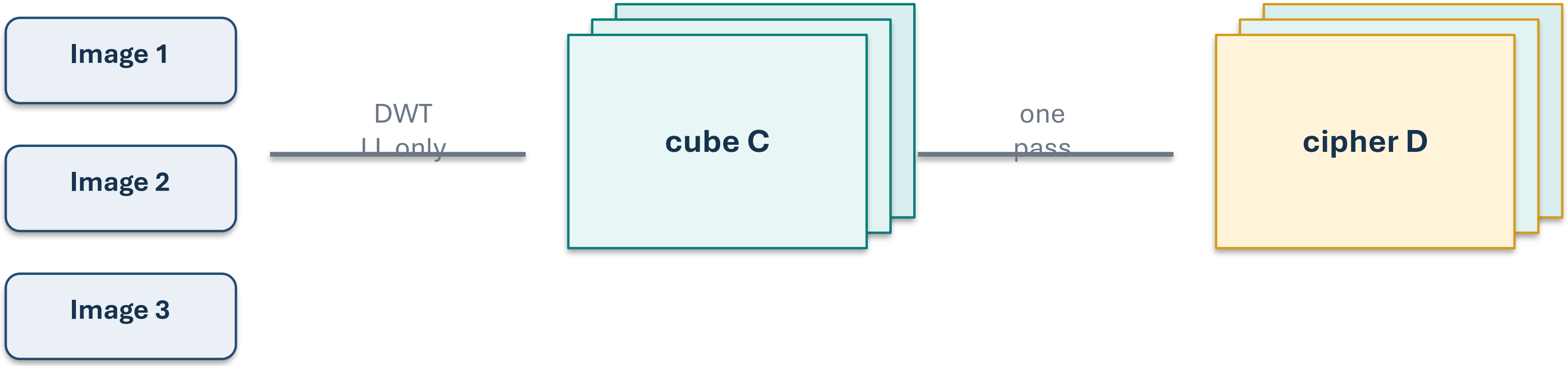
Key idea

Looks random to an attacker, but repeats exactly for the receiver who knows the initial values and parameters.

Ref: Fig. 5 (DWT decomposition) · Fig. 1 (chaotic phase diagram)

Novelty: encrypt the cube, not the images

The novelty is not DWT alone or chaos alone. It is the multi-image cube strategy.



1/4 size

DWT keeps only LL, so each channel becomes half height by half width.

Different sizes

The cube uses the largest dimensions and pads smaller images with zeros.

Single object

Encrypt the whole stack instead of one image at a time.

Ref: Innovations list, Section 1

Methodology: DWT → stack → confusion → diffusion

Abdul → Maria · 1:30 · Method

1

DWT compression

Each R,G,B channel → keep LL only
image shrinks to 1/4 size

2

Stack into cube C

All compressed LL sub-bands
size from largest image

3

Confusion

Chaotic offsets swap pixel positions

4

Diffusion

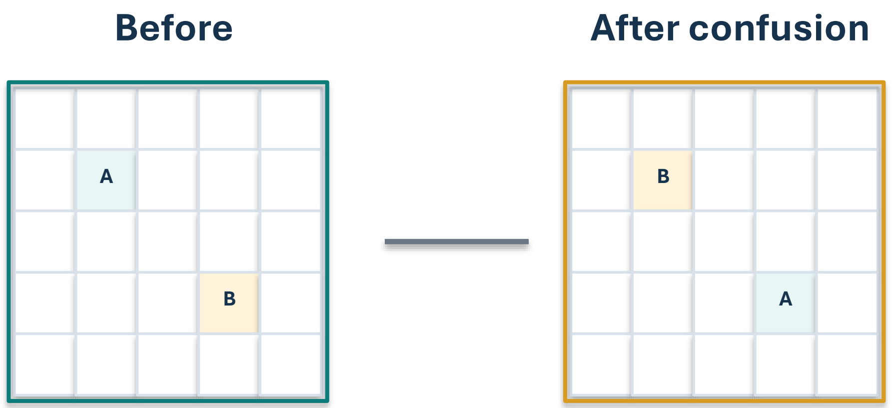
XOR changes values
output cipher cube D

Transmit D. Receiver runs diffusion, confusion, and DWT recovery in reverse.

Ref: Fig. 4 (encryption flow chart) · Fig. 7 (visual results)

Confusion: scrambling pixel POSITIONS

- Iterate chaotic map once per cube pixel.
- Generate coordinate offsets S1, T1, U1.
- Compare each pixel coordinate with its offset.
- Pixel values stay the same; positions move.
- Offsets depend on plaintext, so the scheme is plaintext-aware.



Rule idea: if a coordinate is larger/smaller/equal to its chaotic offset, swap with the corresponding target coordinate. Across i, j, k, this creates nine cases.

Ref: Section 4.1.2 · Equations (8)-(10)

Diffusion: XOR with a chaotic stream

- Flatten C-prime into vector V.
- First pixel XORs with a chaotic seed.
- For later pixels, index mod 3 selects X2, Y2, or Z2.
- Previous encrypted pixel is reused.
- One bit change cascades through the rest.

```
D[k] = V[k]  XOR  chaos[k mod 3]  XOR  D[k-1]
```

V[k] XOR chaos XOR D[k-1] XOR D[k]

Confusion hides where pixels are. Diffusion hides what pixel values are.

Ref: Section 4.1.3 · Equations (11)-(13)

Findings: PSNR 32.06 dB beats prior work

PSNR comparison (dB)

32.0601 dB reconstruction PSNR

1/4 compressed size

6.1529 MB/s encryption speed

Ref: Table 2 (PSNR) · Fig. 7 (visuals) · Table 10 (speed)

Security: NIST · NPCR · UACI · IE all green

Maria · 1:00 · Findings

Security check	Target / meaning	Result	Status
NIST randomness	15 tests	15 / 15 passed	**PASS**
NPCR	target 99.6094%	99.6533%	**PASS**
UACI	target 33.4635%	33.4887%	**PASS**
Information entropy	max 8.0	7.9994	**PASS**
Key space	brute force	huge	**PASS**

What this means

The cipher behaves like noise: flat histogram, low correlation, high sensitivity to tiny plaintext/key changes.

Ref: Tables 1, 5, 6, 8 · Section 6

Limitation and take-home

Maria · 0:30 · Closing

32 dB
PSNR

7.9994
entropy

99.65%
NPCR

Limitation

Zero-padding wastes cube space when image sizes differ. The authors flag this for future work.

Take-home

DWT + chaos gives compression and encryption in one pipeline for multiple color images.

Questions?

Ref: Section 7, Conclusion